

Parametric optimization of pulse-echo laser ultrasonic system for inspection of thick polymer matrix composites

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Abstract

One of the main challenges of using laser ultrasonic techniques for non-destructive testing applications is the typically low signal-to-noise ratio of the laser ultrasonic signals. In the case of thick composite structures, this is even more problematic since composite materials have very strong sound attenuation. This article investigates the effects of laser beam size and profile to the amplitude of pulse-echo laser ultrasonic signals with the constraint that the peak energy density (fluence) must be kept constant under the thermal damage threshold of material like polymer matrix composites. Such constraint is very important for the non-destructive feature of non-destructive testing, yet in a number of the existing parameter studies of laser ultrasonics, it was not fully investigated. In this article, a series of A-scan and C-scan experiments on thick composite specimens shows that the amplitude of the direct waves and the reflected waves increases with the increase in laser beam size with constant peak energy density. This amplitude enhancement significantly improves the propagation depth, thereby optimizing the system for inspection of thick composite structures. The validity of experimental results is verified theoretically by solving the thermoelastic model of epicenter displacement using Laplace–Hankel transformation.

Keywords

Laser thermoelastic model, epicenter displacement, pulse-echo laser ultrasonics, thick composite structures

Introduction

Composite materials are widely used in a number of industrial sectors, including aerospace, marine, and automotive. They demonstrate considerable advantages regarding specific stiffness and specific strength over metallic materials. However, composite materials are vulnerable to a variety of defects, such as disbond and delamination during the manufacturing process or in the service life of the components.¹ To sustain and improve the quality and reliability of composite structures, it is vital to monitor structural health and confirm the system's integrity by identifying the damages before they reach the critical stage.² A number of inspection techniques were proposed for non-destructive testing (NDT) of composite structures, and ultrasonic probe technique is the most commonly used.³ However, this technique has several limitations: the need for an acoustic coupling media, the difficulty of setup, the requirement of skill worker to operate the system, and the requirement of near-normal incidence to the component's surface. Thus, its use is rather limited to

accessible areas, and even inapplicable under harsh and hostile industrial environments.^{4–7}

The laser ultrasonic system uses lasers to generate and detect ultrasonic waves remotely with a number of technical features that able to overcome the limitations of ultrasonic probe technique: non-couplant, non-contact, broad acoustic bandwidth, exceptionally high spatial resolution, and high-speed inspection.^{8–10} For example, the Pulse-Echo Ultrasonic Propagation Imager (PE-UPI) is a noncontact laser-based scanning system providing a series of C-scan wave propagation images for NDT. PE-UPI utilizes the laser ultrasonic

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waves dominated by through-the-thickness longitudinal waves, where the particle motion is parallel to the direction of propagation. It is highly effective at detecting defects at various depths of the structure. For more detailed description of the PE-UIP system, readers can refer to the previous publication.¹¹

One of the main challenges of using laser ultrasonic techniques for NDT applications is the typically low amplitude of the laser ultrasonic signals. In the case of thick composite structures, this is even more problematic since composite materials have very strong sound attenuation. Past and ongoing studies to address this issue can be generally divided into two catalogs: either by spatial modulation techniques that spread the laser energy in space using lenticular lens,¹² optical diffraction gratings,¹³ transmission mask,¹⁴ and laser-induced grating¹⁵ or by temporal modulation techniques that spread the laser energy in time using Bragg cell,¹⁶ pulsed or continuous-wave Q-switching technique,^{17,18} and intensity modulation of diode laser.¹⁹ The key idea of those techniques is to increase the total energy that can be absorbed into the structures without reaching above the thermal damage threshold of the surface's materials. However, those mentioned techniques are greatly exacerbating the complexity and the cost of the NDT system. It is much more preferred to use a simple circular-shaped and single-pulse laser beam, rather than a complex pattern with highly modulated frequency laser beam.

A number of publications were devoted to investigate the effect of laser parameters, such as beam energy, beam size, pulsewidth, and beam profile to the laser ultrasonic signals produced by a circular-shaped and single-pulse laser beam, by analytical solution of thermoelastic models,^{20–24} or modeling using finite element method (FEM),^{25–27} or experimental studies.²⁸ Analytical solutions of the thermoelastic models with different levels of complexity were developed, from early models such as point-source representation,²⁰ surface center of expansion,²¹ to thermoelastic models including thermal diffusion²² and optical penetration.²³ Using analytical solutions, the influence of the laser beam size was studied,²⁴ which shows that the amplitude of the laser-induced acoustic waves increased by decreasing the radius of the focused laser spot. When the model becomes too complicated, for example, temperature-dependent parameters as well as complex geometry, FEM was often used.^{25–27} Wang et al.²⁷ investigated the generation of the elastic waves in non-metallic materials in the thermoelastic regime using FEM, taking into account thermal diffusion, finite spatial and temporal shape of the laser pulse, optical penetration, and temperature dependence of material properties. Experimental studies were often used for verification of theoretical results. The experiment with

polymer-matrix composites of Dubois et al.²⁸ showed that the laser-generated amplitude decreases for long pulse duration, but the effect is more significant at high frequencies.

Given the extended existing literatures, one can find that the solution to increase the signal amplitude of the laser ultrasonics is by increasing the beam energy or by reducing the beam size. However, this solution misses to consider the non-destructive feature of the NDT techniques, which is that laser energy density must be kept under the thermal damage threshold of materials, which is typically quite low in case of polymer matrix composites.²⁹ When the energy density constraint is considered, the optimization process for laser parameters is no longer a single variable optimization but a compromise of several factors including beam energy, beam size, and beam profile. In this article, we investigate the effect of laser beam size on the pulse-echo ultrasonic wave with the constraint of constant peak energy density. These results of this study lead to a more efficient and reliable technique using laser ultrasonics for inspection of thick composite structures.

This article is organized as follows. In the next section, a thermoelastic model of epicenter displacement by laser pulse is presented, following by a series of experiments A-scan and C-scan laser ultrasonics of an 8.0 mm thickness and a 10.3 mm thickness composite specimen. Finally, this article concludes with section "Conclusion."

Thermoelastic model of epicenter displacement by pulse laser

Formulation of thermoelastic model

Figure 1 shows the schematic of the epicenter displacement generated by a short pulse laser and measured using an optical sensing beam. The thermoelastic process is governed by the coupling equations of thermoelasticity^{22,23}

$$\nabla^2 T - \kappa^{-1} \dot{T} - c_H^{-2} \ddot{T} = -k^{-1} W \quad (1)$$

$$(\lambda + 2\mu) \nabla \phi - 2\mu \nabla \times \psi = \rho \ddot{u} + \gamma \nabla T \quad (2)$$

$$W = E(1 - R_i)h(z)g(t)f(r) \quad (3)$$

$$h(z) = \gamma_e e^{-\gamma_e z} \quad (4)$$

$$g(t) = \frac{t}{\sqrt{2}} e^{-t/\nu} \quad (5)$$

$$f(r)_{\text{super-Gaussian}} = \frac{1}{\sqrt{2\pi}} \frac{2}{R_G} e^{-2r^n/R_G^n} \quad (6)$$

where κ is the thermal diffusivity, $c_H = \sqrt{\kappa/\tau}$ is the heat wave speed, τ is the material relaxation time, k is the thermal conductivity, W is the heat source power

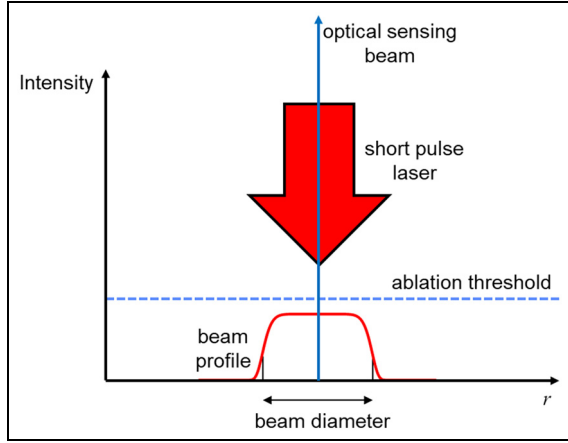


Figure 1. Epicenter displacement produced by a short pulse laser in thermoelastic regime and measured using an optical sensing beam.

density, λ and μ are the Lamé coefficients, φ and ψ are the dilatational and rotational potentials, ρ is the density, $\gamma = (3\lambda + 2\mu)\alpha_T / (\lambda + 2\mu)$ is the thermoacoustic coupling constant, α_T is the volumetric thermal expansion coefficient, T is the temperature rise above ambient, E is the pulse energy, R_i is the surface reflectivity, $h(z)$ is the absorption profile, $g(t)$ is the temporal profile, $f(r)$ is the spatial profile (super-Gaussian profile with order n), γ_e is the extinction coefficient, ν is the laser pulse rise time, and R_G is $1/e^2$ radius of the Gaussian beam.

The system of equations (1) to (6) is solved using the Laplace–Hankel transformation. The solution of out-of-plane displacement at the surface $z = 0$ in transformed domain is²³

$$\tilde{u}_z|_{z=0} = \gamma b^2 k^{-1} s^2 (\beta^2 + p^2) \left[\frac{\kappa}{s} (\alpha \xi^{-1} - 1) + \frac{1}{\xi^2 - \gamma_e^2} (\alpha \gamma_e^{-1} - 1) \right] \frac{\gamma_e^2}{\gamma_e^2 - \xi^2} R^{-1} \tilde{W} \quad (7)$$

where $\alpha^2 = a^2 s^2 + p^2$, $\beta^2 = b^2 s^2 + p^2$, $\xi^2 = \alpha^2 + s/\kappa$, $R = (\beta^2 + p^2)^2 - 4\alpha\beta p^2$, $\tilde{W} = W_0 W_p(p) W_s(s)$, $W_0 = E(1 - R_i)$, $W_p(p) = \mathcal{H}[f(r)]$, $W_s(s) = \mathcal{L}((t/\nu^2)e^{-t/\nu}) = 1/(vs + 1)^2$; $a = c_1^{-1}$ and $b = c_2^{-1}$, c_1 and c_2 are the longitudinal and transverse speeds, respectively, $\mathcal{L}()$ and $\mathcal{H}()$ are the Laplace and Hankel transformations, respectively.

Finally, the out-of-plane displacement at the surface $z = 0$ is calculated by the inverse Laplace–Hankel transform of equation (7)

$$u_z|_{z=0} = \frac{1}{2\pi i} \int_0^{\infty} \int_{\sigma_p - i\infty}^{\sigma_p + i\infty} p J_0(pr) e^{st} \tilde{u}_z|_{z=0} ds dp \quad (8)$$

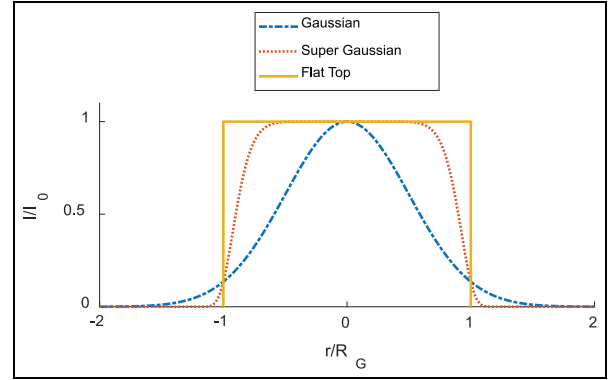


Figure 2. Spatial profile of Gaussian, super-Gaussian order 10, and flat-top laser beam; I_0 is peak energy density and R_G is the $1/e^2$ radius of the laser beam.

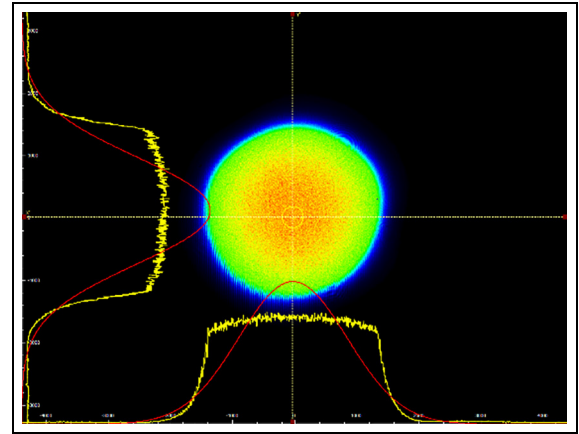
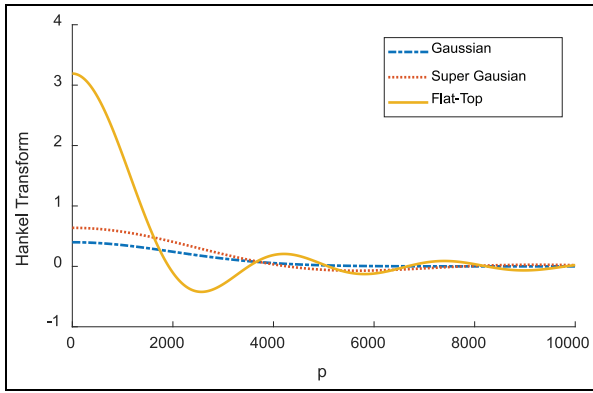


Figure 3. Laser profile beam with beam radius of 1.58 mm corresponding to beam area 7.80 mm². The yellow lines in the figure are the measured intensity distributions in horizontal and vertical directions across the center beam.

The model presented above has two notable differences with the original model of Spicer.²² First, the optical penetration effect was included with the finite extinction coefficient γ_e . Spicer's solution²² is a special case of equation (7) where the extinction coefficient approaches infinite. Second, the spatial profile is super-Gaussian instead of the commonly used Gaussian profile (Figure 2). The reason the super-Gaussian profile was considered is that the actual measured laser profile of the laser used in experiment fits with the super-Gaussian order 10 (Figure 3). Even though the analytical Hankel transform of super-Gaussian profile is not available, its Hankel transform can be calculated by direct integration. Figure 4 shows the Hankel transformation of Gaussian, super-Gaussian, and flat-top profiles.

Table 1. Material parameters used in the numerical study.

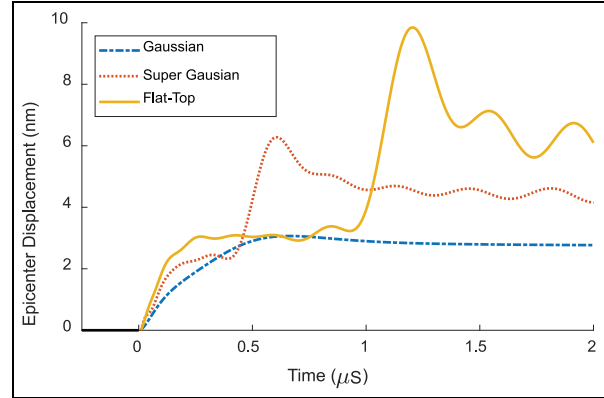
Parameters	Notation	Unit	Numerical value
Surface reflectivity	R_i	—	0.75
Thermal expansion coefficient	α_T	K^{-1}	10^{-6}
Thermal diffusivity	κ	$m^2 s^{-1}$	10^{-6}
Thermal conductivity	k	$W m^{-1} K^{-1}$	0.2
Longitudinal velocity	c_1	$m s^{-1}$	3000
Transverse velocity	c_2	$M s^{-1}$	2000
Thermoacoustic coupling	γ	K^{-1}	1.2×10^{-6}
Extinction coefficient	γ_e	m^{-1}	0.7×10^7

**Figure 4.** Hankel transforms of Gaussian, super-Gaussian order 10, and flat-top profiles.

Numerical study

In this section, we present the numerical study with the thermoelastic model presented in previous section. The material parameters are shown in Table 1 corresponding to the typical values of carbon fiber–reinforced polymer (CFRP) composites.³⁰ The values used for the laser parameters are peak laser energy density $I_0 = 370 \text{ J m}^{-2}$ and laser pulse rise time $\nu = 8 \times 10^{-9} \text{ s}$.

Effect of laser beam profile to epicenter displacement with the constraint of constant peak energy density. A parametric study is carried out to investigate the influence of the laser beam profile with the constraint of constant peak energy density. Three profile cases are studied: Gaussian, super-Gaussian with order $n = 10$, and flat-top profiles with the laser beam radius of 0.93 mm. The beam energy of each profile can be calculated using integration in equation (9). The laser beam with flat-top profile has the highest energy, following by super-Gaussian profile, and the Gaussian profile beam has the lowest energy

**Figure 5.** Epicenter displacement with different laser profiles, beam radius of 0.93 mm, and peak energy density of 370 J m^{-2} .

$$E = 2\pi I_0 \int_0^{\infty} f(r) dr \quad (9)$$

The calculated displacement epicenter waveforms are shown in Figure 5. This result shows that the epicenter displacement at the “tail” of the waveform has the largest amplitude with flat-top profile beam, followed by super-Gaussian profile, and finally by Gaussian profile beam. Considering that in all three cases the peak energy density is the same, this result shows that flattening the laser profile is effective to improve laser ultrasonic signals while ensuring no ablation occurs.

Effect of laser beam size to epicenter displacement with the constraint of constant peak energy density. In this section, we present the result of the numerical study to investigate the effect of laser beam size to the epicenter displacement with the constraint of constant peak energy density. The laser profile is super-Gaussian with order $n = 10$ and the material properties are shown in Table 1. The laser beam radius R_G are 0.93, 1.25, and 1.58 mm, corresponding to the beam area of 2.69, 4.90, and 7.80 mm^2 , and the beam energy of 1.00, 1.82, and 2.89 mJ, respectively.

The calculated displacement epicenter waveforms are shown in Figure 6. Several observations can be derived as follows. The early time displacements represent the quick local thermal expansion associated with the direct heating. At longer times, the effect of lateral extent of the source becomes apparent with the arrival of the surface waves. By enlarging the source, the arrival of the surface waves is delayed where the peak of the waveform shifts to the right of the time axis. The effect of the laser profile also can be observed in Figure 6 where the waveform after the peak

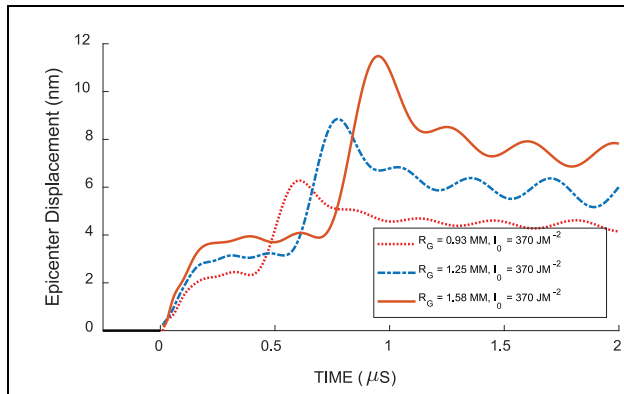


Figure 6. Epicenter displacement with different beam sizes and constant peak energy density, and the laser profile is super-Gaussian with order 10.

oscillates, unlike the smooth waveform in Spicer and Hurley's²² paper, where a Gaussian laser profile was assumed. Finally, the effect of laser beam size is clearly shown: the epicenter displacement at the "tail" of the waveform increases when laser beam radius increases. Considering that the peak energy density is the same in all three presented cases, it is concluded that using a larger beam size is effective to improve laser ultrasonic signals while ensuring no ablation of material occurs.

A-scan and C-scan experiments

Experiment setup

The A-scan and C-scan experiments were carried out to investigate the effect of laser beam size and laser energy density to the pulse-echo laser ultrasonics on polymer matrix composites. Two samples of CFRP composite specimens were provided by Boeing Research & Technology as shown in Figure 7(a) and (b). Two specimens were made from the same type of material, but different in thickness: the specimen in Figure 7(a) has the thickness of 8.0 mm and the specimen in Figure 7(b) has the thickness of 10.3 mm. In both specimens, there were PZT sensors remaining from previous experiments that are considered here as simulated damages with added masses. In addition, the specimen with the thickness of 10.3 mm has two impact damages.

Figure 8 shows the A-scan experiment setup. A 1064 nm Q-switched Nd:YAG laser was used for generating the thermoelastic ultrasonic waves. A 633 nm laser Doppler vibrometer (LDV) single-point sensor was used for sensing. The sensitivity of the LDV was set to $10 \text{ mm s}^{-1} \text{ V}^{-1}$, which allows the signal to be captured with a frequency of up to 250 kHz. A digitizer with the sampling frequency of 59.35 MHz was used for data

acquisition. The signal from the LDV was filtered with the range from 0.10 to 0.30 MHz. An optical assembly was made to combine the laser ultrasonic generation and sensing beams. In order to obtain the optimum signal-to-noise ratio, the sensing beam was set normal to the test sample at the maxima visibility standoff distance 642 mm; also, the experiment was repeated 15 times and averaged. Before the test, no special surface treatment was made, only simple cleaning using soft cloth.

Figure 9 shows the C-scan experiment using the PE-UPI system. Only one-time scanning was performed for the C-scan experiment due to the very large number of points in the scan area. For the 8.0-mm-thick composite specimen, the scan area was $390 \text{ mm} \times 80 \text{ mm}$ (Figure 6); for the 10.3-mm-thick composite specimen, the scan area was $520 \text{ mm} \times 300 \text{ mm}$ (Figure 7). The scan interval was set at 0.25 mm for both cases. The scan speed was set at 1.8 kHz, which means that the system can scan 1800 points per second. The scanning time for the 8.0-mm-thick composite specimen was 9.6 min; the scanning time for the 10.3-mm-thick composite specimen was 42 min at the scan interval of 0.25 mm.

A-scan, B-scan, and C-scan representations of laser ultrasonic wavefield

The ultrasonic wavefield acquired by the PE-UPI system has four typical representations: A-scan, B-scan, and C-scan as shown in Figure 10, and tomographic video representation of the C-scan results. A-scan displays a single ultrasonic wave as a function of time. B-scan displays the cross section of the three-dimensional wavefield by sliding along a scanning line. C-scan displays the plane perpendicular to the time axis, which gives an image of the internal structure of the test object in plan-view. Successive slicing by the C-scan representation and displaying at an appropriate video rate provides the pulse-echo ultrasonic wave propagation video in time domain.

In pulse-echo mode, a typical A-scan signal can be divided into two parts: the direct wave is the immediate epicenter displacement of the surface due to direct laser heat, and the reflected wave includes multiple wave reflected back-and-forth between two surfaces of the structure (Figure 11). Typically, the direct waves have a large amplitude, which can easily mask the early reflected waves of the first few cycles. For NDT purpose, the reflected waves are more important since it penetrates deep inside the structures. Higher amplitude of reflected waves means that thicker specimen can be inspected using laser ultrasonic system.

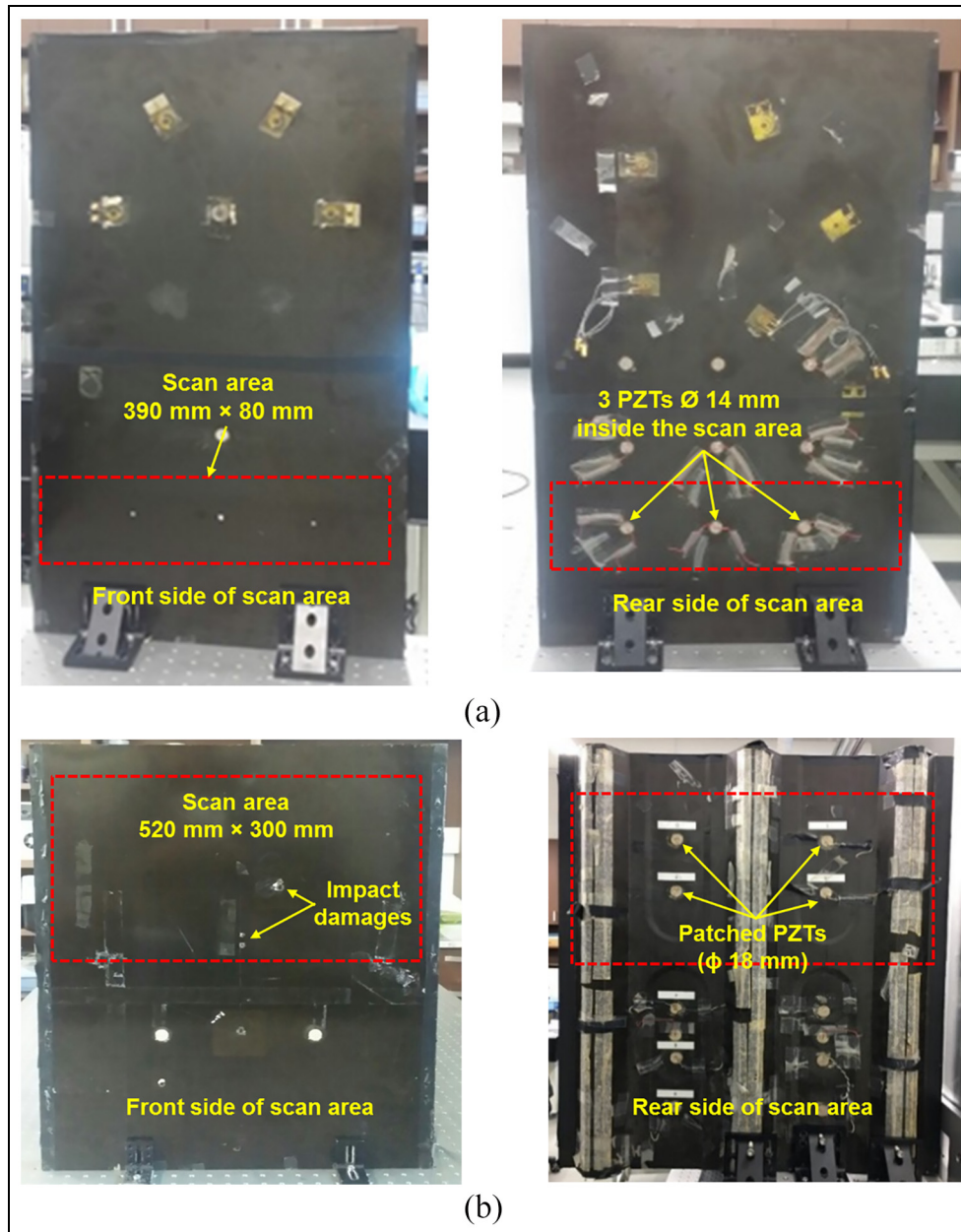


Figure 7. CFRP composite specimens with PZT attachments at the rear side of the scan area (a) 8.0 mm thick and (b) 10.3 mm thick. The specimen with the thickness of 10.3 mm has two impact damages.

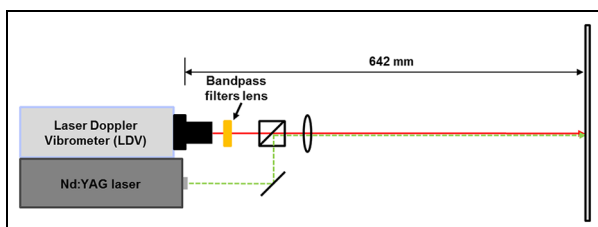


Figure 8. A-scan experiment setup to measure the epicenter pulse-echo laser ultrasonics.

Effect of laser beam size to the A-scan laser ultrasonics

Figure 12(a) and (b) shows the A-scan pulse-echo ultrasonic signals of the 8.0-mm and 10.3-mm-thick CFRP composite specimens, respectively. The laser energy was set to 1.00, 1.82, and 2.89 mJ, with the corresponding laser beam areas at the impinging point of 2.69, 4.90, and 7.80 mm², respectively. In each case, the energy density was maintained at 370 J m⁻². It is

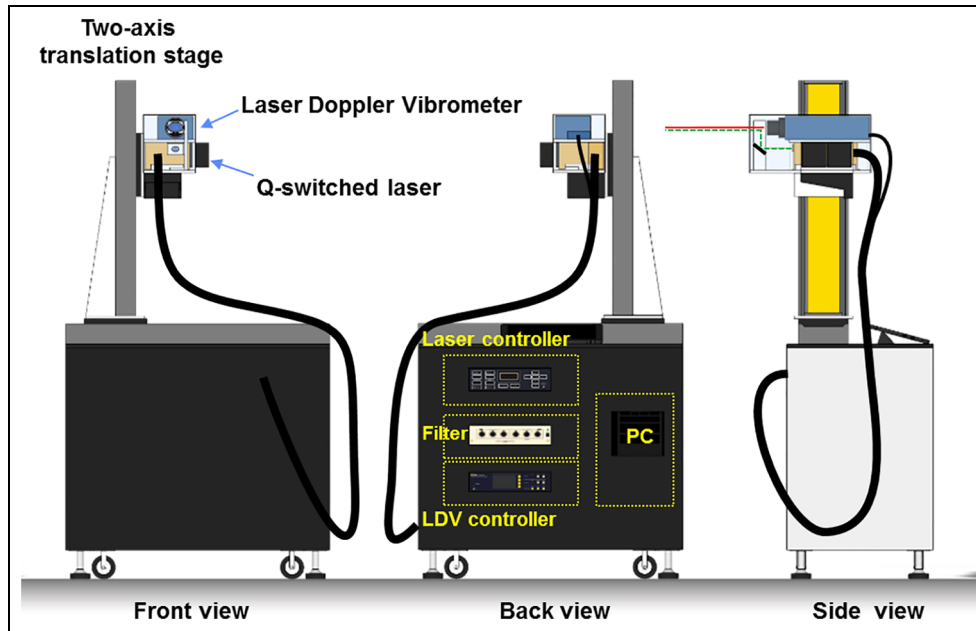


Figure 9. C-scan experiment using the Pulse-Echo Ultrasonic Propagation Imager (PE-UPI) system.

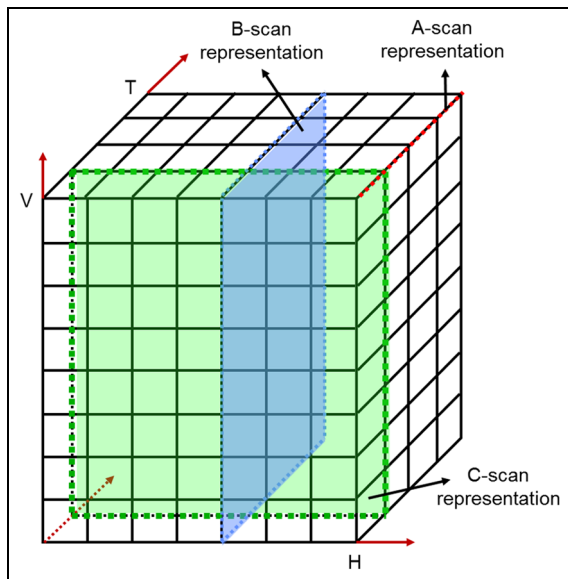


Figure 10. A-scan, B-scan, and C-scan representations of the laser ultrasonic wavefield.

shown that the amplitude of the signals increases when the laser beam size increases as predicted in section “Thermoelastic model of epicenter displacement by pulse laser.” It is worth noting that the experimental waveforms are different from theoretical waveforms in section “Thermoelastic model of epicenter displacement by pulse laser” mainly due to the low sampling frequency of the sensing device (in the paper of Spicer and

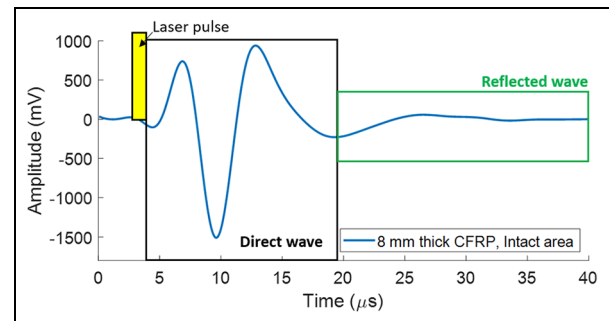


Figure 11. Typical A-scan ultrasonic signal in pulse-echo mode of the PE-UPI system in 8-mm-thick CFRP composite specimen. Direct waves is the immediate epicenter displacement of the surface due to direct laser heat, and the reflected waves includes multiple wave reflected back-and-forth between two surfaces of the structure.

Hurley,²² a sensor with bandwidth of 80 MHz was used).

Figure 13(a) and (b) plots the peak-to-peak amplitude of the direct waves and the reflected waves of the A-scan signals. The black square dots represent the case of the 8.0-mm-thick composite specimen and the red circle dots represent the case of the 10.3-mm-thick composite specimen. The bottom horizontal axis in Figure 13 shows the laser beam area (2.69, 4.90, and 7.80 mm²) corresponding to the top horizontal axis which shows the laser beam energy (1.00, 1.82, and 2.89 mJ). It is again emphasized that the peak energy

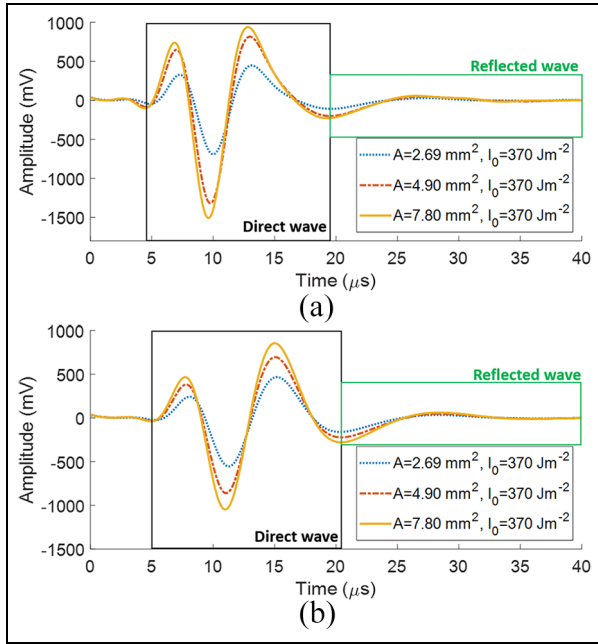


Figure 12. A-scan pulse-echo mode ultrasonic signals of the (a) 8.0-mm-thick and (b) 10.3-mm-thick CFRP composite specimen at the same peak energy density of 370 J m^{-2} with different laser beam sizes.

density was constant at 370 J m^{-2} . The vertical axis is the amplitude of the ultrasonic signals.

The results in Figure 13(a) and (b) clearly show the influence of laser beam size on the strength of the reflected and direct waves. The peak-to-peak amplitude of the reflected waves of the 8.0-mm specimen increases

by 74.3% as the laser beam size increases from 2.69 to 4.90 mm^2 . Similarly, the peak-to-peak amplitude of the reflected waves of the 10.3-mm specimen increases by 52.0% as the laser beam size increases from 2.69 to 4.90 mm^2 . This reflected wave amplitude enhancement significantly increases the propagation depth, which leads to an enhancement of the C-scan images presented in the next section.

Effect of laser beam size to C-scan damage visualization images

Figure 14(a) to (c) shows the C-scan images of the 8.0-mm-thick CFRP composite specimen. Those images were obtained from the freeze frames of the pulse-echo ultrasonic wave propagation videos by the mobile laser PE-UPI system (M_PE_UPI system: L type, X-NDT Inc.).³¹ The laser energy was set to 1.00, 1.82, and 2.89 mJ, with the corresponding laser beam areas at the impinging point 2.69, 4.90, and 7.80 mm^2 , respectively. In each case, the peak energy density was maintained at 370 J m^{-2} , far below the thermal damage threshold of material.

The 14-mm-diameter PZTs patched at the rear side of the 8.0-mm-thick CFRP specimen were able to visualize in the case of using 4.90 and 7.80 mm^2 laser beam area in Figure 14(b) and (c). However, it could not be visualized in the case of using 2.69 mm^2 laser beam area (Figure 14(a)). This observation is consistent with the A-scan results in previous section where a larger laser beam with constant energy density leads to enhancements in laser ultrasonics amplitude.

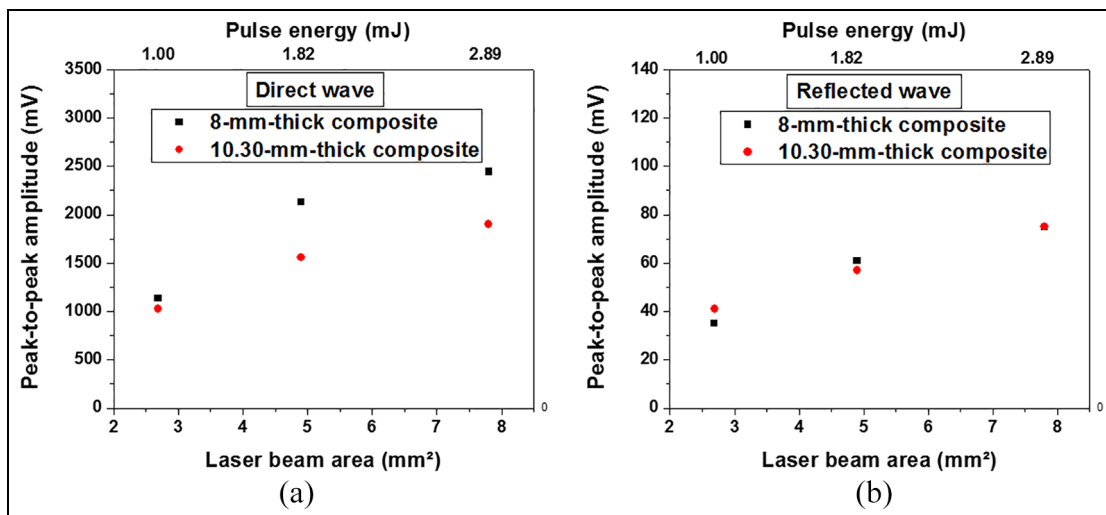


Figure 13. Peak-to-peak amplitude of the A-scan pulse-echo mode ultrasonic signals of the 8.0-mm and 10.3-mm-thick CFRP composite specimen for (a) direct waves and (b) reflected waves.

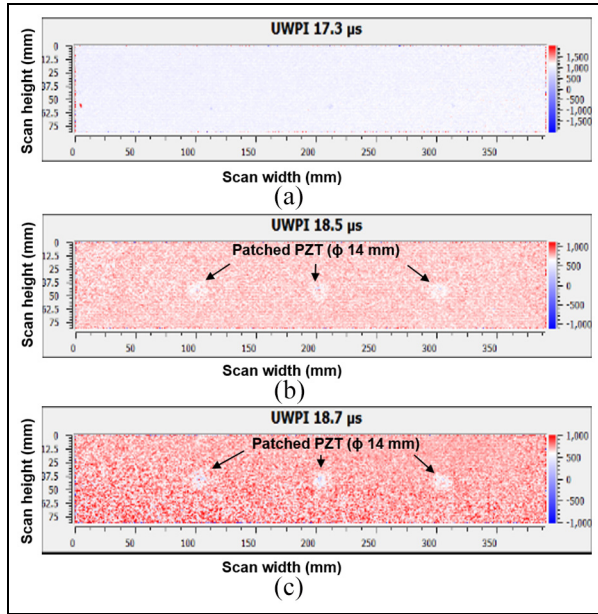


Figure 14. C-scan pulse-echo mode ultrasonic images of the 8.0-mm-thick CFRP composite specimen at the same peak energy density of 370 J m^{-2} with different laser beam areas: (a) 2.69 mm^2 , (b) 4.90 mm^2 , and (c) 7.80 mm^2 .

Performance improvement with high energy density

As observed in Figure 14(a), the patched PZTs on the rear side of the 8.0-mm-thick CFRP composite specimen could not be visualized at 370 J m^{-2} energy density with 2.69 mm^2 laser beam area. In this section, we attempt to improve the detection performance by increasing the energy density of the laser to a value nearer but still below the thermal damage threshold of composite material.

First, the experiment to measure the thermal damage threshold of the composite specimen was performed. The damage appears as a bleached spot on the surface of the CFRP composite specimen, which appears around the energy density of 1000 J m^{-2} . Based on this value, the C-scan experimental parameters were set as follows: the generation laser was set to 2.00, 3.64, and 5.80 mJ pulse energy, and the corresponding laser beam size at the impinging point was 2.69 , 4.90 , and 7.80 mm^2 , respectively. In each case, the fluence remained constant at 744 J m^{-2} , well below the thermal damage threshold.

Figure 15(a) to (c) shows the C-scan images of the 8.0-mm-thick CFRP composite specimen with constant energy density at 744 J m^{-2} . For all three cases, the patched PZTs that simulate added mass on the rear side can be observed, even in the case of using laser

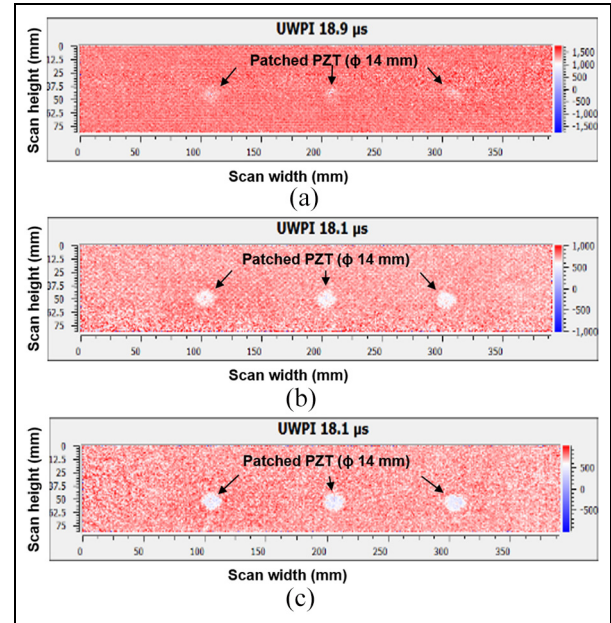


Figure 15. C-scan pulse-echo mode ultrasonic images of the 8.0-mm-thick CFRP composite specimen at the same peak energy density of 744 J m^{-2} with different laser beam laser beam areas: (a) 2.69 mm^2 , (b) 4.90 mm^2 , and (c) 7.80 mm^2 .

beam area 2.69 mm^2 (Figure 15(a)). In addition, one can easily find an improvement of the detectability of the added mass with the increase in laser beam area.

Optimized laser parameters for inspection of the 10.3-mm-thick composite specimen

The study of laser parameters in previous sections is applied for C-scan inspection of the 10.3-mm-thick composite specimen with two impact damages. Previous C-scan inspection attempts of this specimen using a highly focused laser beam leads to burning of the surface with poor C-scan results.

First, laser parameters are selected. Given that the ablation threshold of material is approximately 1000 J m^{-2} , the peak energy density was chosen at 744 J m^{-2} . Next, the laser pulse energy was chosen at 3.64 mJ, which leads to the laser beam area of 4.90 mm^2 . The scan area was $520 \text{ mm} \times 300 \text{ mm}$ with scan interval was 0.25 mm as shown in Figure 7(b). Figure 16 shows the C-scan pulse-echo mode ultrasonic images at 9.8 and $15.1 \mu\text{s}$. The impact damages can be visualized clearly in both figures as well as the three PZTs of 18 mm diameter in the rear side of the specimen. The other PZT was not visualized due to internal defect of structure. The internal structure of the specimen can also be visualized.

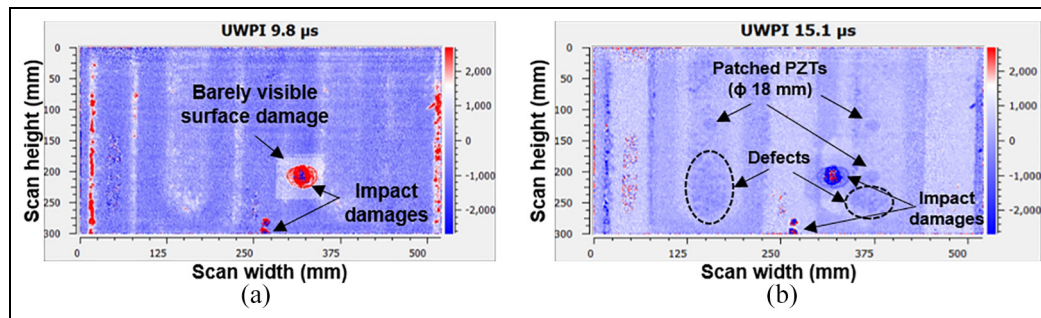


Figure 16. C-scan pulse-echo mode ultrasonic images of the of the 10.3-mm-thick CFRP composite specimen at (a) 9.80 μs and (b) 15.10 μs pulse-echo ultrasonic wave propagation imaging video freeze frame.

Conclusion

To the present day, thick composite structures still present a significant challenge to NDT techniques using laser ultrasonics due to the low thermal damage threshold as well as extremely high sound attenuation of composite materials. This article studies how laser beam size affects inspection performance of pulse-echo laser ultrasonics on CFRP composites given the constraint that laser energy density must be kept under the thermal damage threshold of materials, for example, a polymer matrix composite. Experiments were performed to study the amplitude of the A-scan signals and C-scan images with different combinations of laser pulse energies and laser beam sizes on CFRP composite materials. It is shown from experimental results that a larger laser beam generates greater peak-to-peak amplitudes of the direct waves and reflected waves with the constant energy density. This amplitude enhancement significantly increases the propagation depth of the ultrasonic waves into the structure and therefore improves the capability to inspect thick composite structures.

The experimental results were verified with a numerical study of the thermoelastic model of epicenter displacement. The presented model has two notable differences with the original model of Spicer and Hurley.²² First, the optical penetration effect was included, represented by the finite value of extinction coefficient γ_e . Second, a realistic laser profile was used instead of commonly used Gaussian profile. The results of the numerical study showed that flattening the laser profile is effective to improve laser ultrasonic signals while ensuring no ablation occurs; also, it confirms the experiment results where the epicenter displacement at the “tail” of the waveform increases when laser beam radius increases given constant energy density.

Declaration of conflicting interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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